

Sunya Afrasiabi

email | personal website | linkedin

Education

University of Pennsylvania , BSE in Mechanical Engineering, BA in Philosophy	Sept 2024 - May 2028
• Relevant Coursework: Thermal-Fluids Engineering, Statics & Strength of Materials, Thermodynamics, Dynamics, Computer-Aided Design (CAD), Computer Programming (Python), Linear Algebra, Differential Equations, Probability	
University at Buffalo , Gifted Math Program	Sept 2023 - April 2024
• Relevant Coursework: College Calculus III, Introduction to Linear Algebra	

Experience & Leadership

Munch Industries, Inc. , Mechanical Engineer	July 2026 - Present
• Designed and fabricated the awning and structural framework for Orble, an automated boba tea vending machine startup, using SolidWorks, 3D printing, and PVC fabrication. Assisted with material selection and procurement, and contributed to engineering and operational decisions during product development.	
University of Pennsylvania – Li Group , Undergraduate Researcher	May 2026 - Present
• Research insect-wing-inspired morphing structures, modeling compliant geometries in Rhino and Grasshopper to study parametric mechanical behavior. • Translate computational designs into prototypes using powder-based 3D printing for experimental validation.	
NASA L'SPACE Proposal Writing & Evaluation Experience , Radiation Subteam Lead	Sept 2025 - Dec 2025
• Developed innovative solution to the limitation of radiation-resistant materials in HDPE composites to enhance material performance under lunar radiation conditions • Gained experience in technical proposal composition, project evaluation, and participated in NASA review panel. • Ranked in the Top 6 out of 40+ national teams.	
Penn Astronomical Student Association , Outreach & Marketing Director	Sept 2025 - Present
• Lead outreach initiatives and external partnerships, securing \$200 in sponsorship funding. • Configure and calibrate equatorial telescopes and optical tracking systems for public observation events.	
#MyStory Program , Founder and Project Manager	Jan 2021 - Aug 2024
• Founded and managed a non-profit oral history initiative promoting cultural awareness through exhibitions. • Secured over \$24,000 via targeted donations, fundraisers, and corporate grant programs (including Riley's Way Foundation, Prudential, Ashoka, and the Peter & Elizabeth Tower Foundation). • Directed operations for 20+ community partners and 40+ volunteers.	

Technical Projects

Multi-Phase Butane Rocket & Numerical Analysis	Spring 2026
• Designed and optimized a variable-mass butane-powered water rocket using aerodynamic modeling, thermodynamic analysis, and experimental validation to navigate constrained flight paths and minimize target error. • Developed a Python Euler integration solver coupling butane vaporization, pressure-driven propulsion, drag, and ballistic dynamics; implemented trajectory reconstruction to improve predictive accuracy.	
Mars Rover Chassis Prototype	Spring 2026
• Designed and optimized a Mars rover chassis prototype for University Rover Challenge (URC) constraints as part of a 3-person engineering team, developing detailed SolidWorks assemblies and structural architectures. • Performed FEA-driven design iterations to optimize load paths, bracket placement, wall thicknesses, and lightweight geometries while balancing structural integrity and mass reduction.	
Vacuum-Powered Agricultural Seed Cleaner	Fall 2025
• Designed and fabricated a vertical pneumatic seed-chaff separator with a zig-zag separation column, custom swirl chamber, laser-cut acrylic panels, and 3D-printed nozzles to optimize airflow behavior and separation efficiency.	

Papers

Linear Algebra Foundations of Neural Networks , University of Pennsylvania	August 2025
• Wrote expository research paper connecting linear algebra, neural networks, and robotics. (Link)	
Aviation Solutions for Wildlife Prevention , Wharton Undergraduate Aerospace Club	May 2025
• Co-authored investor-oriented whitepaper on UAVs, AI wildfire prediction, and aerospace innovations, framing \$6.4B+ market opportunities and potential \$5B annual cost savings. (Link)	

Skills

Software: Rhino 3D, Grasshopper, SolidWorks, MATLAB, Python, NumPy, FEA
Fabrication: Laser Cutting, FDM/SLA 3D Printing, Mechanical Assembly
Additional: Computational Geometry, Bio-Inspired Design, Technical Writing, Logic, Project Management